

**Cancer Prevention**  
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Cancer is fundamentally a disease of accumulated genetic mutations, and aging is the primary driver of that accumulation. Several biological mechanisms converge in older adulthood to elevate cancer risk:

- Genomic instability: Over decades, cellular DNA accumulates replication errors, oxidative damage, and inadequate repair. This cumulative genomic instability underlies the majority of age-related cancers.
- Immunosenescence: The aging immune system becomes progressively less effective at identifying and destroying malignant cells, a process called immune surveillance decline.
- Telomere attrition: Telomeres — the protective caps on chromosomes — shorten with each cell division. Critically short telomeres promote chromosomal instability and senescence, which can paradoxically increase cancer risk through the senescence-associated secretory phenotype.
- Epigenetic drift: Age-related changes in DNA methylation and histone modification alter gene expression, sometimes silencing tumor-suppressor genes.
- Chronic low-grade inflammation (inflammaging): A persistent state of subclinical systemic inflammation, common in older adults, creates a microenvironment conducive to tumor initiation and progression.

*Most Common Cancers in Older Adults*

| Cancer Type          | Median Age at Diagnosis | Key Modifiable Risk Factors                 |
|----------------------|-------------------------|---------------------------------------------|
| Prostate             | 66                      | Diet, physical activity                     |
| Breast               | 62                      | Alcohol, obesity, physical inactivity       |
| Lung                 | 70                      | Tobacco, radon exposure                     |
| Colorectal           | 66                      | Diet, obesity, physical inactivity, alcohol |
| Bladder              | 73                      | Tobacco, occupational chemicals             |
| Non-Hodgkin Lymphoma | 67                      | Immune function, obesity                    |
| Skin (non-melanoma)  | 65+                     | UV exposure, prior sunburns                 |
| Endometrial          | 60                      | Obesity, estrogen exposure                  |
| Pancreatic           | 70                      | Tobacco, obesity, diabetes                  |

*The Case for Prevention at Any Age*

A common misconception is that cancer prevention is primarily relevant for younger adults. However, research consistently shows that lifestyle modifications adopted in later life still yield significant protective benefit. For example:

- Smoking cessation at age 65 still reduces lung cancer risk by approximately 50% over 10 years compared to continued smoking.
- Increased physical activity in older adults is associated with reduced risk of colon, breast, and endometrial cancers regardless of when the activity is initiated.
- Adoption of a Mediterranean-style diet is associated with reduced cancer incidence and all-cause mortality even when begun after age 60.

*Tobacco Cessation*

Tobacco use remains the single most preventable cause of cancer death worldwide. Smoking is causally linked to at least 15 cancer types, including cancers of the lung, larynx, oropharynx, esophagus, stomach, kidney, bladder, cervix, and acute myeloid leukemia (IARC, 2023). Among older adults, many of whom began smoking decades ago, the carcinogenic burden from tobacco exposure

is substantial. Smokeless tobacco and e-cigarettes are not safe alternatives. Smokeless tobacco is directly linked to oral, esophageal, and pancreatic cancers. While the long-term carcinogenic profile of e-cigarettes is not yet fully established, their vapor contains known carcinogens including formaldehyde and acrolein. Cessation is effective at all ages. The following evidence-based approaches are available:

Pharmacological:

- Nicotine Replacement Therapy (NRT): Available as patches, gum, lozenges, nasal spray, and inhalers. Doubles cessation success rates compared to placebo.
- Varenicline (Chantix/Champix): A partial nicotinic agonist that significantly reduces cravings. Most effective pharmacological option but requires monitoring for cardiovascular and psychiatric side effects in some individuals.
- Bupropion (Zyban): An atypical antidepressant that reduces cravings. Contraindicated in individuals with seizure disorders.

Behavioral:

- Individual or group counseling
- Telephone quitlines (e.g., 1-800-QUIT-NOW in the US)
- Cognitive behavioral therapy (CBT)-based programs

Combination approaches (NRT + behavioral counseling) yield the highest long-term cessation rates. Older adults may have more entrenched habits but also stronger motivation (existing health conditions, desire to spend time with grandchildren). Physiological changes in drug metabolism may alter medication dosing needs. Consultation with a primary care physician is strongly advised before initiating pharmacotherapy.

### Dietary Strategies for Cancer Prevention

No single food prevents cancer; rather, overall dietary patterns matter most. The American Cancer Society (ACS), World Cancer Research Fund (WCRF), and American Institute for Cancer Research (AICR) converge on the following evidence-based dietary guidance:

- Emphasize plant foods: vegetables, fruits, whole grains, and legumes
- Limit red and processed meat
- Avoid ultra-processed foods and sugar-sweetened beverages
- Minimize foods high in refined carbohydrates and added sugars
- Moderate alcohol consumption

Vegetables in the Brassicaceae family — including broccoli, cauliflower, Brussels sprouts, kale, cabbage, and arugula — contain glucosinolates, which are metabolized into bioactive compounds (isothiocyanates such as sulforaphane) that:

- Induce Phase II detoxification enzymes in the liver
- Promote apoptosis (programmed cell death) in precancerous cells
- Inhibit angiogenesis (formation of tumor blood vessels)
- Exhibit anti-inflammatory properties

### *Colorful Fruits and Vegetables*

A diverse range of fruits and vegetables supply antioxidants, carotenoids (beta-carotene, lycopene, lutein), polyphenols, and flavonoids that combat oxidative stress, a major driver of DNA damage and carcinogenesis.

- Tomatoes (cooked): High in lycopene, associated with reduced prostate cancer risk
- Berries: Rich in anthocyanins and ellagic acid; associated with reduced esophageal, colon, and oral cancer risk
- Dark leafy greens (spinach, chard): Rich in folate, essential for DNA repair
- Citrus fruits: Contain limonene and vitamin C; associated with reduced risk of esophageal and stomach cancer

#### *Whole Grains and Dietary Fiber*

Dietary fiber — particularly from whole grains, legumes, and vegetables — is strongly and inversely associated with colorectal cancer risk. Proposed mechanisms include:

- Dilution and acceleration of intestinal transit, reducing carcinogen contact time with colonic mucosa
- Fermentation by gut microbiota producing short-chain fatty acids (SCFAs), notably butyrate, which promotes colonocyte health and apoptosis of malignant cells
- Reduction of circulating insulin and insulin-like growth factor-1 (IGF-1)

#### *Legumes*

Beans, lentils, chickpeas, and peas are high in fiber, protein, phytochemicals, and resistant starch. Regular legume consumption is associated with reduced risk of colorectal and prostate cancer.

#### *Fatty Fish and Omega-3 Fatty Acids*

Eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), found in fatty fish such as salmon, sardines, mackerel, and herring, exert anti-inflammatory effects that may mitigate cancer risk, particularly for colorectal and breast cancers. Recommended intake: At least 2 servings (3.5 oz each) of fatty fish per week.

#### *Green Tea*

Green tea contains catechins — particularly epigallocatechin gallate (EGCG) — which have demonstrated antiproliferative and proapoptotic effects in laboratory studies. Epidemiological evidence, particularly from East Asian populations, associates regular green tea consumption with reduced risk of gastric, esophageal, and colorectal cancers.

#### *Vitamin D and Calcium*

Vitamin D deficiency is common among older adults and has been associated with increased risk of colorectal, breast, and prostate cancers. Older skin is less efficient at synthesizing vitamin D from sunlight, making dietary sources and supplementation more critical.

- Dietary sources: Fortified milk and orange juice, fatty fish, egg yolks, mushrooms exposed to UV light
- Supplementation: Many older adults benefit from 800–2,000 IU/day of vitamin D<sub>3</sub>; consult a physician for individualized guidance
- Calcium (from diet or supplements, 1,000–1,200 mg/day) is associated with reduced colorectal polyp recurrence

#### Foods and Patterns to Limit

##### *Red and Processed Meat*

The World Health Organization (WHO) classifies processed meat (e.g., bacon, sausage, hot dogs, deli meats) as a Group 1 carcinogen (causally linked to colorectal cancer) and red meat as Group 2A (probably carcinogenic). The mechanisms involve:

- Heme iron catalyzing formation of N-nitroso compounds and lipid peroxidation products in the colon
- Heterocyclic amines (HCAs) and polycyclic aromatic hydrocarbons (PAHs) generated during high-temperature cooking (grilling, frying)
- Sodium nitrites used as preservatives in processed meats

Recommendation: Limit red meat to no more than 12–18 oz (cooked weight) per week; minimize processed meat consumption to rare occasions.

### *Ultra-Processed Foods*

Ultra-processed foods — defined as industrial formulations containing ingredients not typically found in home cooking (emulsifiers, colorants, artificial flavors, high-fructose corn syrup) — have been associated in multiple large prospective cohort studies with increased overall cancer risk. They are typically calorie-dense, fiber-poor, and nutritionally inadequate.

### *Salt and Salt-Preserved Foods*

High dietary sodium, particularly from salted, pickled, or smoked foods, is associated with increased gastric cancer risk. The World Cancer Research Fund recommends limiting sodium to less than 2,400 mg per day.

### *Dietary Patterns*

Overall dietary patterns associated with reduced cancer risk include:

- Mediterranean diet: High in olive oil, fish, vegetables, legumes, nuts, and whole grains
- DASH diet: Designed for blood pressure management but also associated with reduced cancer risk
- Plant-forward (flexitarian) diet: Predominantly plant-based with modest animal product inclusion

## Physical Activity and Exercise

Physical activity is one of the most robustly established modifiable risk factors for cancer prevention. The AICR and ACS identify physical inactivity as contributing to cancers of the colon, breast (postmenopausal), endometrium, kidney, esophagus, and stomach. Regular exercise reduces cancer risk through multiple mechanisms:

- Reduction of circulating estrogen and androgens
- Improvement of insulin sensitivity and reduction of IGF-1 levels
- Reduction of systemic inflammation (lower CRP, IL-6, TNF- $\alpha$ )
- Enhancement of immune surveillance and NK cell activity
- Promotion of healthy body weight and adiposity reduction
- Facilitation of healthy bowel motility (reducing colorectal cancer risk)

The U.S. Department of Health and Human Services (2018 Physical Activity Guidelines for Americans) recommends that older adults achieve:

| Activity Type              | Weekly Target    | Examples                                              |
|----------------------------|------------------|-------------------------------------------------------|
| Moderate-intensity aerobic | ≥150–300 minutes | Brisk walking, swimming, cycling, water aerobics      |
| Vigorous-intensity aerobic | ≥75–150 minutes  | Jogging, hiking uphill, aerobics classes              |
| Muscle-strengthening       | ≥2 days/week     | Resistance bands, light weights, bodyweight exercises |
| Balance/flexibility        | ≥3 days/week     | Tai chi, yoga, stretching                             |

For those unable to meet these targets due to chronic conditions or disability, any amount of physical activity is better than none. Sedentary time should be broken up throughout the day, even with short movement breaks.

### *Aerobic Exercise*

Regular moderate-intensity walking (30 minutes most days of the week) is the most accessible and studied form of aerobic exercise in older adults. A meta-analysis by Friedenreich et al. (2021) found that meeting aerobic activity guidelines was associated with 10–20% reduction in risk for several cancer types compared to being sedentary. Water-based exercise (aqua aerobics, swimming) is particularly valuable for older adults with arthritis, osteoporosis, or joint pain, as it provides aerobic conditioning with minimal impact stress.

### *Resistance Training*

Muscle mass declines with age (sarcopenia), leading to metabolic consequences including insulin resistance. Resistance training counteracts this by:

- Improving glucose metabolism and insulin sensitivity
- Reducing visceral adiposity
- Improving immune function

Two to three sessions per week targeting major muscle groups (upper body, lower body, core) using weights, resistance bands, or bodyweight are recommended.

### *Mind-Body Exercise*

Practices such as yoga, tai chi, and qigong offer dual benefits of physical conditioning and stress reduction. They are particularly well-suited to older adults given their focus on balance, flexibility, and low-impact movement. Emerging evidence suggests stress reduction itself may reduce cancer risk through neuroendocrine-immune pathways.

### *Reducing Sedentary Behavior*

Independent of structured exercise, prolonged sitting is associated with increased cancer risk.

Strategies to reduce sedentary time include:

- Using a timer to stand and walk for 2–5 minutes every 30–60 minutes
- Standing desks or adjustable workstations
- Walking during phone calls
- Parking farther away or getting off public transit one stop early
- Replacing screen time with active hobbies (gardening, walking clubs)

### Alcohol Consumption

Alcohol is classified as a Group 1 carcinogen by the International Agency for Research on Cancer (IARC). It is causally linked to cancers of the oral cavity, pharynx, larynx, esophagus, liver, colorectum, and breast. The mechanisms include:

- Acetaldehyde (a toxic metabolite of ethanol) directly damaging DNA
- Generation of reactive oxygen species (oxidative stress)
- Impaired folate metabolism, increasing genomic instability
- Increased circulating estrogen levels (promoting hormone-receptor-positive breast cancer)

Critically, there is no established safe threshold for alcohol consumption with respect to cancer risk. Risk increases in a dose-dependent manner beginning with the very first drink. The notion that moderate alcohol consumption (one drink per day) is protective against cancer has been consistently refuted in recent large-scale analyses.

- Ideal: Complete abstinence from alcohol eliminates this carcinogenic exposure entirely
- If drinking: The ACS (2020) updated guidelines recommend that for cancer prevention, it is best not to drink alcohol at all
- For those who choose to drink, current conservative limits are no more than 1 drink/day for women and 2 drinks/day for men — but these limits were established for cardiovascular, not cancer, risk

One standard drink is defined as 14 grams of pure alcohol: approximately 12 oz of regular beer (5% ABV), 5 oz of wine (12% ABV), or 1.5 oz of 80-proof spirits.

### *Special Considerations for Older Adults*

Alcohol metabolism changes with age — the same quantity of alcohol results in higher blood alcohol concentrations in older adults compared to younger adults. This increases both the carcinogenic exposure per drink and the risk of falls and drug interactions. Many older adults take medications (anticoagulants, metformin, anticonvulsants) that interact adversely with alcohol.

### Sun Safety and UV Exposure

Skin cancer is the most commonly diagnosed cancer in the United States, and incidence increases markedly with age due to cumulative UV exposure over a lifetime. The three major types are:

- Basal cell carcinoma (BCC): Most common; rarely metastasizes but can cause significant local tissue damage
- Squamous cell carcinoma (SCC): More aggressive; can metastasize if untreated
- Melanoma: Least common but most deadly; strongly associated with intense, intermittent UV exposure and prior sunburns

Ultraviolet radiation (both UVA and UVB) damages DNA in keratinocytes and melanocytes by inducing pyrimidine dimers, which, if unrepaired, lead to mutations in tumor-suppressor genes (particularly TP53 and CDKN2A).

### *Protective Strategies*

- Sunscreen: Apply broad-spectrum SPF 30 or higher to all exposed skin 15–30 minutes before sun exposure. Reapply every 2 hours, or after swimming or sweating. Mineral sunscreens (zinc oxide, titanium dioxide) are preferred for sensitive older skin.
- Protective clothing: Long sleeves, wide-brimmed hats (>3 inch brim), and UV-protective clothing (UPF 50+) are more reliable than sunscreen alone.
- UV-protective sunglasses: Protect against ocular melanoma and reduce periorbital skin exposure.
- Time of day: Avoid sun exposure between 10 AM and 4 PM, when UV radiation is most intense.
- Shade: Seek shade whenever possible; note that sand and water reflect UV and can increase exposure.

- Tanning beds: Completely avoid. Indoor tanning devices emit concentrated UVA radiation and are classified as Group 1 carcinogens. They increase melanoma risk by 75% when first used before age 35 (Boniol et al., 2012).

### *Skin Self-Examination*

Monthly self-examination of the skin is recommended using the ABCDE rule for evaluating moles and skin lesions:

| Letter | Criterion | Warning Sign                                     |
|--------|-----------|--------------------------------------------------|
| A      | Asymmetry | Two halves do not match                          |
| B      | Border    | Irregular, ragged, or blurred edges              |
| C      | Color     | Multiple shades of brown, black, red, or blue    |
| D      | Diameter  | Larger than 6 mm (pencil eraser)                 |
| E      | Evolving  | Any change in size, shape, color, or new symptom |

Any suspicious lesion should be evaluated promptly by a dermatologist.

### Sleep and Circadian Health

Epidemiological evidence increasingly links sleep disruption, circadian rhythm misalignment, and insufficient sleep duration with elevated cancer risk. The primary mechanisms involve:

- Melatonin suppression: Light exposure at night suppresses melatonin production. Melatonin has oncostatic properties: it inhibits cancer cell proliferation, suppresses estrogen synthesis, and acts as a direct antioxidant. Night shift workers — who chronically disrupt their circadian rhythm — have elevated risks of breast, colorectal, and prostate cancers (IARC, 2019).
- Immune dysregulation: Sleep deprivation reduces natural killer (NK) cell activity and T-cell function, impairing immune surveillance.
- HPA axis dysregulation: Chronic sleep disruption elevates cortisol, which suppresses immune function and promotes a pro-inflammatory state.
- Obesity promotion: Insufficient sleep increases appetite-regulating hormones (ghrelin rises, leptin falls), promoting overeating and obesity — itself a major cancer risk factor.

Sleep duration: Older adults should aim for 7–9 hours of sleep per night. Those consistently sleeping fewer than 6 or more than 9 hours show elevated disease risk.

Sleep hygiene practices:

- Maintain a consistent sleep and wake time, even on weekends
- Keep the bedroom cool, dark, and quiet
- Avoid screens (phones, tablets, televisions) for at least 1 hour before bed; blue light suppresses melatonin
- Avoid caffeine after 2 PM
- Avoid alcohol within 3 hours of bedtime (disrupts sleep architecture despite promoting initial drowsiness)
- Engage in a calming pre-sleep routine (reading, light stretching, meditation)
- Limit naps to 20–30 minutes and avoid late afternoon napping

Sleep disorders: Sleep apnea, restless legs syndrome, and insomnia are highly prevalent in older adults and should be evaluated and treated. Untreated sleep apnea causes recurrent hypoxia and oxidative stress that may independently elevate cancer risk.

Environmental and Occupational Exposures

*Radon*

Radon is a naturally occurring radioactive gas produced by uranium decay in soil and rock. It is the second leading cause of lung cancer after tobacco, responsible for approximately 21,000 deaths per year in the US. It seeps into homes through foundation cracks, particularly in basements.

Action: Test your home for radon. Test kits are inexpensive and widely available. If levels exceed 4 picocuries per liter (pCi/L), mitigation is recommended (typically sub-slab depressurization systems). Radon is invisible and odorless — testing is the only way to detect it.

*Asbestos*

Asbestos fibers, when inhaled, cause mesothelioma and lung cancer. Many older adults were occupationally exposed (construction, shipbuilding, automotive, textile industries) prior to widespread regulations. Older homes may contain asbestos in insulation, floor tiles, or ceiling materials.

Action: Do not disturb asbestos-containing materials. Hire a certified professional for inspection and abatement if renovation work is planned. If occupational exposure history is significant, discuss with your physician.

*Air Pollution*

Outdoor fine particulate matter (PM2.5) and indoor air pollution are recognized carcinogens for lung cancer. Older adults with limited mobility may spend more time indoors, where air quality may be poor due to combustion (gas stoves, wood fireplaces) or volatile organic compounds (VOCs) from paints and cleaning products. Recommendations:

- Monitor local air quality (AirNow.gov in the US) and limit outdoor activity on high-pollution days
- Use exhaust ventilation when cooking with gas
- Use low-VOC paints and cleaning products
- Consider a HEPA air purifier in living areas

*Pesticides and Chemicals*

Chronic exposure to certain pesticides, herbicides, and industrial chemicals has been associated with lymphomas, leukemias, and other cancers. Older adults who worked in agriculture, manufacturing, or who used residential pesticides extensively may have accumulated significant exposures.

Recommendations: Wash produce thoroughly; consider organic produce for the "Dirty Dozen" (EWG annual list). Avoid unnecessary use of household pesticides.

Preventive Screenings and Vaccinations

Cancer screening does not prevent cancer per se but enables detection at earlier, more treatable stages, meaningfully reducing cancer mortality. For some cancers, pre-cancerous lesions can be identified and removed before malignant transformation occurs (e.g., colorectal polyps).

The following recommendations are based on U.S. Preventive Services Task Force (USPSTF), American Cancer Society, and National Comprehensive Cancer Network (NCCN) 2024 guidelines. Individual circumstances — comorbidities, life expectancy, patient preferences — should be discussed with a physician to determine appropriateness.

| Cancer     | Screening Method | Recommended Ages                          | Frequency                                   |
|------------|------------------|-------------------------------------------|---------------------------------------------|
| Colorectal | Colonoscopy      | 45–75 (45–85 with shared decision making) | Every 10 years                              |
| Colorectal | FIT or Cologuard | 45–75                                     | Annually (FIT); every 1–3 years (Cologuard) |

|                |                            |                                                     |                 |
|----------------|----------------------------|-----------------------------------------------------|-----------------|
| Breast (women) | Mammography                | 40-74                                               | Every 1–2 years |
| Cervical       | Pap + HPV co-test          | Up to age 65 (if adequate prior screening)          | Every 5 years   |
| Lung           | Low-dose CT (LDCT)         | 50–80, ≥20 pack-year history, current/recent smoker | Annually        |
| Prostate       | PSA test (men)             | 55–69 (shared decision making)                      | Every 1–4 years |
| Skin           | Full-body dermatology exam | 60+, high-risk individuals                          | Annually        |

Note: Screening decisions for individuals over 75 should be individualized based on functional status, life expectancy, and patient values. In individuals with limited life expectancy, the harms of screening may outweigh the benefits.

### ***Vaccinations***

Several vaccines directly prevent infection-associated cancers:

| Vaccine               | Cancers Prevented                                              | Recommendation for Older Adults                                         |
|-----------------------|----------------------------------------------------------------|-------------------------------------------------------------------------|
| Hepatitis B (HBV)     | Hepatocellular (liver) cancer                                  | Recommended for unvaccinated adults aged 19–59; shared decision for 60+ |
| HPV vaccine           | Cervical, oropharyngeal, anal, vulvar, vaginal, penile cancers | Approved through age 45; discuss with physician if older and at risk    |
| Hepatitis C treatment | Liver cancer (not a vaccine; antiviral therapy)                | Screen once for HCV; curative treatment is available                    |

### **Psychosocial Wellbeing and Stress**

Chronic psychological stress activates the hypothalamic-pituitary-adrenal (HPA) axis and sympathetic nervous system, producing sustained elevations in cortisol and catecholamines (epinephrine, norepinephrine). These stress hormones have multiple downstream effects relevant to cancer:

- Suppression of NK cell cytotoxicity and T-lymphocyte function
- Promotion of systemic inflammation (NF-κB activation)
- Induction of angiogenesis via beta-adrenergic receptor signaling on tumor cells
- Behavioral effects: stress-induced insomnia, poor dietary choices, increased alcohol use, and physical inactivity compound risk

### ***Social Isolation and Loneliness***

Social isolation and loneliness are highly prevalent among older adults and have been associated with accelerated immunosenescence, increased inflammation, and poorer cancer survival outcomes. Maintaining robust social connections is not merely a quality-of-life concern — it has documented biological health implications.

### ***Evidence-Based Stress Management Strategies***

| Strategy                                  | Evidence Level      | Notes                                             |
|-------------------------------------------|---------------------|---------------------------------------------------|
| Mindfulness-Based Stress Reduction (MBSR) | High (RCT evidence) | Reduces cortisol, improves NK cell activity       |
| Cognitive Behavioral Therapy (CBT)        | High                | Effective for anxiety, depression, insomnia       |
| Regular aerobic exercise                  | High                | Potent anxiolytic and antidepressant effect       |
| Yoga and tai chi                          | Moderate            | Reduces inflammation markers, improves mood       |
| Social engagement                         | Moderate            | Community groups, volunteering, family connection |
| Pet ownership                             | Moderate            | Associated with reduced stress biomarkers         |
| Nature exposure                           | Moderate            | "Forest bathing" (shinrin-yoku) reduces cortisol  |
| Gratitude journaling                      | Low-moderate        | Emerging evidence for mood and immune benefits    |

### *Addressing Depression and Anxiety*

Clinical depression and anxiety in older adults are underdiagnosed and undertreated. Both conditions are independently associated with poorer cancer outcomes, likely through immune dysregulation and behavioral mechanisms. Screening for depression (PHQ-9) and anxiety (GAD-7) should be incorporated into routine primary care visits. Effective treatments include psychotherapy, pharmacotherapy, and combined approaches.

### Weight Management and Metabolic Health

Excess body fat — particularly visceral adiposity — is a well-established risk factor for at least 13 cancer types, including cancers of the esophagus, stomach, colon, rectum, liver, gallbladder, pancreas, kidney, endometrium, ovary, breast (postmenopausal), thyroid, and multiple myeloma. The biological mechanisms linking obesity to cancer include:

- Hyperinsulinemia and elevated IGF-1: Promote cell proliferation and inhibit apoptosis
- Adipose-derived estrogen: Adipose tissue is a major site of estrogen synthesis in postmenopausal women; excess adipose tissue drives estrogen-dependent cancers
- Adipokine dysregulation: Obesity increases leptin (mitogenic) and decreases adiponectin (anti-inflammatory, pro-apoptotic)
- Chronic inflammation: Adipose tissue macrophage infiltration creates a pro-inflammatory, pro-tumorigenic microenvironment

Weight management in older adults is nuanced: while obesity is a cancer risk factor, unintentional weight loss (cachexia) in older adults carries its own serious health risks, including muscle wasting, immunosuppression, and frailty. The goal is healthy weight maintenance rather than aggressive caloric restriction. Evidence-based strategies:

- Focus on dietary quality (nutrient density) rather than calorie counting
- Prioritize protein intake (1.0–1.2 g/kg body weight/day) to preserve muscle mass
- Regular resistance training to maintain lean body mass
- Avoid rapid or extreme dietary interventions without medical supervision
- Monitor for unintentional weight loss, which may require evaluation for underlying malignancy